

Callum Huntington

callumhuntington@gmail.com · callum.huntington@unina.it
callumhuntington.github.io

Education

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|  PhD in Mathematics and its Applications
University of Naples Federico II, Italy | 2023–present |
|  MSc in Mathematics (with Distinction)
University of Sheffield, United Kingdom | 2020–2021 |
|  BSc in Mathematics (with Class One Honours)
University of Sheffield, United Kingdom | 2017–2020 |

Research Experience

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|  PhD Researcher
University of Naples Federico II, Italy

Research focus: spectral theory of signed graphs and gain graphs, with emphasis on structural operations (joins, products, corona constructions) and switching equivalence. Developed spectral methods using linear algebra and group representations for gain graphs over arbitrary groups.

Collaborations with researchers at Niccolò Cusano University, Italy (S. Spessato, A. Donno, M. Cavaleri), University of Belgrade, Serbia (Z. Stanić), University of the Western Cape, South Africa (O. Waka), and Eindhoven University of Technology, The Netherlands (A. Abiad, A. Khramova). | 2023–present |
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Publications

Published

-  **C. Huntington.** Spectra and balance of signed graphs obtained from vertex splittings. *Discussiones Mathematicae Graph Theory*, in press, 2026. DOI: 10.7151/dmgt.2643.
-  **C. Huntington, S. Spessato, F. Belardo.** On H -joins of complex unit gain graphs and their stability. *Discrete Mathematics*, vol. 349, no. 7, pp. 115069, 2026. DOI: 10.1016/j.disc.2026.115069.
-  **C. Huntington.** The H -join operation of signed graphs constrained by indexing maps. *RAIRO-Operations Research*, vol. 59, no. 5, pp. 2501–2515, 2025. DOI: 10.1051/ro/2025100.

-  F. Belardo, **C. Huntington**. Signed graphs with strong (anti-)reciprocal eigenvalue property. *Special Matrices*, vol. 12, no. 1, pp. 20240017, 2024. DOI: 10.1515/spma-2024-0017.

Under review and in preparation

-  **C. Huntington**, Z. Stanić, F. Belardo. Further spectral relations between oriented graphs and corresponding signed graphs. *Under review*.
-  **C. Huntington**, O. Waka. Spectra of gain Q and R graphs via G -phases and group representations. *Under review*.
-  **C. Huntington**, F. Belardo, S. Barik. Graphs with integer reciprocal eigenvalue properties. *Under review*.
-  F. Belardo, A. Donno, **C. Huntington**, S. Spessato. A switching-stable corona product of signed graphs. *Under review*.
-  **C. Huntington**. Spectral properties and balance of products derived from gain subdivision graphs via group representations. *In preparation*.
-  A. Abiad, **C. Huntington**, A.P. Khramova. A switching method for signed and gain sun graphs. *In preparation*.
-  **C. Huntington**. Quotient graphs and spectra of Leonardo grids. *In preparation*.

Conferences

-  **2nd Joint Meeting Brazil–Italy in Mathematics** September 2026
Messina, Italy
invited talk: Graphs and signed graphs with integer reciprocal eigenvalue properties
-  **Real-world Applications of Geometry and Algebra 2026** June 2026
Eindhoven, The Netherlands
-  **Combinatorics 2026** May 2026
Naples, Italy
-  **Mini Workshop of Topology and Algebra 2025** December 2025
Cape Town, South Africa
invited talk: The H -join of complex unit gain graphs and their stability
-  **Workshop on Spectral Graph Theory 2025** October 2025
Gramado, Brazil
contributed talk: Spectral relations between oriented graphs and corresponding signed graphs

-  **12th PhD Summer School in Discrete Mathematics**
Koper, Slovenia
contributed talk: The H -join of complex unit gain graphs and their stability

September 2025
-  **CIME Summer School: Modern Perspectives in Approximation Theory**
Cetraro, Italy

July 2025
-  **IRGTADS Seminars**
Online
invited talk: The H -join operation on signed graphs

June 2025
-  **Group Theory in Salerno**
Salerno, Italy

March 2025
-  **1st Chinese–SouthEastEuropean Conference on Discrete Mathematics and Applications**
Belgrade, Serbia
contributed talk: Constructions of nonregular cospectral signed graphs

June 2024
-  **University of Pisa Summer School: Introduction to Machine Learning in Geosciences**
Pisa, Italy

July 2022

Awards and Funding

-  **Travel Grant for RAGA 2026**
Eindhoven University of Technology – €715

2026
-  **Doctoral Programme Scholarship**
University of Naples Federico II – €16,243

2023–2026

Additional Experience

-  **Warehouse Operative**
Arthrex UK, Sheffield, United Kingdom

2016–2023, 2026–present

Other Skills and Qualifications

-  **Languages:** English (native), Italian (B1).
-  **Programming:** Basic knowledge of Python, SageMath, R, and MATLAB.
-  **Refereeing activity:** Served as a peer reviewer for several international journals in graph theory and discrete mathematics.